SATGURU STUDY CENTRE

Chennai - 81

TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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Class : 11

COMMON QUARTERLY EXAMINATION - 2024 - 25

BUSINESS MATHEMATICS AND STATISTICS

Time Allowed : 3.00 Hours] **[Max. Marks : 90**

PART - I

20 x 1 = 20

1. Answer all the questions by choosing the correct answer from the given 4 alternatives
2. Write question number, correct option and corresponding answer
3. Each question carries 1 mark

1. The value of $\begin{vmatrix} 2x+y & x & y \\ 2y+z & y & z \\ 2z+x & z & x \end{vmatrix}$ is

(a) xyz (b) $x+y+z$ (c) $2x+2y+2z$ (d) 0

2. The number of Hawkins-Simon conditions for the viability of an input-output analysis is

(a) 1 (b) 3 (c) 4 (d) 2

3. If A is an invertible matrix of order 2 then det (A^{-1}) be equal to

(a) det (A) (b) $\frac{1}{\det(A)}$ (c) 1 (d) 0

4. If $A = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$, then $|2A|$ is equal to

(a) $4\cos 2\theta$ (b) 4 (c) 2 (d) 1

5. The value of n, when $np_2 = 20$ is

(a) 3 (b) 6 (c) 5 (d) 4

6. For all $n > 0, nC_1 + nC_2 + nC_3 + \dots + nC_n$ is equal to

(a) 2^n (b) $2^n - 1$ (c) n^2 (d) $n^2 - 1$

7. The value of $(5C_0 + 5C_1) + (5C_1 + 5C_2) + 1\,(5C_2 + 5C_3) + (5C_3 + 5C_4) + (5C_4 + 5C_5)$ is

(a) $2^6 - 2$ (b) $2^5 - 1$ (c) 2^8 (d) 2^7

8. The number of ways to arrange the letters of the word " CHEESE"

(a) 120 (b) 240 (c) 720 (d) 6

9. If m_1 and m_2 are the slopes of the pair of lines given by $ax^2 + 2hxy + by^2 = 0$, then the value of $m_1 + m_2$ is

(a) $\frac{2h}{b}$ (b) $-\frac{2h}{b}$ (c) $\frac{2h}{a}$ (d) $-\frac{2h}{a}$

10. The locus of the point P which moves such that P is always at equidistance from the line $x + 2y + 7 = 0$ is

(a) $x + 2y + 2 = 0$ (b) $x - 2y + 1 = 0$ (c) $2x - y + 2 = 0$ (d) $3x + y + 1 = 0$

11. $ax^2 + 4xy + 2y^2 = 0$ represents a pair of parallel lines then 'a' is

(a) 2 (b) -2 (c) 4 (d) -4

12. Equation of the directrix of the parabola $y^2 = -x$

(a) $4x + 1 = 0$ (b) $4x - 1 = 0$ (c) $x - 4 = 0$ (d) $x + 4 = 0$

13. The value of $\sin(-420^\circ)$ is

(a) $\frac{\sqrt{3}}{2}$ (b) $-\frac{\sqrt{3}}{2}$ (c) $\frac{1}{2}$ (d) $\frac{-1}{2}$

14. The value of $\sec A \sin(270^\circ + A)$ is

(a) -1 (b) $\cos^2 A$ (c) $\sec^2 A$ (d) 1

15. $\sec^{-1}\frac{2}{3} + \operatorname{cosec}^{-1}\frac{2}{3} =$

(a) $\frac{-\pi}{2}$ (b) $\frac{\pi}{2}$ (c) π (d) $-\pi$

16. The value of $\frac{1}{\operatorname{cosec}(-45^\circ)}$ is

(a) $\frac{-1}{\sqrt{2}}$ (b) $\frac{1}{\sqrt{2}}$ (c) $\sqrt{2}$ (d) $-\sqrt{2}$

17. If $f(x) = \begin{cases} x^2 - 4x & \text{if } x \geq 2 \\ x + 2 & \text{if } x < 2 \end{cases}$, then $f(0)$ is

(a) 2 (b) 5 (c) -1 (d) 0

18. Which of the following function is neither even nor odd?

(a) $f(x) = x^3 + 5$ (b) $f(x) = x^5$ (c) $f(x) = x^{10}$ (d) $f(x) = x^2$

19. If $y = x$ and $z = \frac{1}{x}$, then $\frac{dy}{dz} =$

(a) x^2 (b) 1 (c) $-x^2$ (d) $-\frac{1}{x^2}$

20. $\frac{d}{dx}(a^x) =$

(a) $\frac{1}{x \log_e a}$ (b) a^a (c) $x \log_e a$ (d) $a^x \log_e a$

PART - II

1. Answer any 7 questions **7 x 2 = 14**
2. Each question carries 2 marks
3. Question number 30 is compulsory

21. Solve $\begin{vmatrix} x-1 & x & x-2 \\ 0 & x-2 & x-3 \\ 0 & 0 & x-3 \end{vmatrix} = 0$.

22. Show that $\begin{bmatrix} 8 & 2 \\ 4 & 3 \end{bmatrix}$ is non - singular.

23. If $15C_{3r} = 15C_{r+3}$, find r.

24. How many distinct words can be formed using all the letters of the following words : MISSISSIPPI

25. Find the distance of the point (4,1) from the line $3x - 4y + 12 = 0$.

26. Find the equation of the circle when the end points of the diameter are (2,4) and (3, − 2).

27. Prove that $\tan^{-1}\left(\frac{4}{3}\right) - \tan^{-1}\left(\frac{1}{7}\right) = \frac{\pi}{4}$.

28. Find the values of each of the following trigonometric ratios. : $\cos(-105°)$.

29. Evaluate: $\lim_{n\to\infty} \frac{\Sigma n^2}{n^3}$.

30. If $y = 2 + \log x$, then show that $xy_2 + y_1 = 0$.

PART - III

1. Answer any 7 questions **7 x 3 = 21**
2. Each question carries 3 marks
3. Question number 40 is compulsory

31. Solve: $\begin{vmatrix} 7 & 4 & 11 \\ -3 & 5 & x \\ -x & 3 & 1 \end{vmatrix} = 0$.

32. Find n, if $\frac{1}{9!} + \frac{1}{10!} = \frac{n}{11!}$.

33. If a polygon has 44 diagonals, find the number of its sides.

34. If $A(-1,1)$ and $B(2,3)$ are two fixed points, then find the locus of a point P so that the area of triangle APB = 8 sq.units.

35. If the lines $3x - 5y - 11 = 0$, $5x + 3y - 7 = 0$ and $x + ky = 0$ are concurrent, find the value of k.

36. Find the angle between the straight lines $x^2 + 4xy + y^2 = 0$.

37. If $\sin A = \frac{3}{5}$, find the values of $\cos 3A$ and $\tan 3A$.

38. Evaluate: $\lim_{x\to 1} \frac{x^3-1}{x-1}$.

39. Find $\frac{dy}{dx}$: $x^2 - xy + y^2 = 7$

40. If $\lim_{x\to 2} \frac{x^n-2^n}{x-2} = 448$, then find the least positive integer n.

PART - IV

1. Answer all the questions **7 x 5 = 35**
2. Each question carries 5 marks

41. a) If $X = \begin{bmatrix} 8 & -1 & -3 \\ -5 & 1 & 2 \\ 10 & -1 & -4 \end{bmatrix}$ and $Y = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 2 & 1 \\ 5 & p & q \end{bmatrix}$ then, find p, q if $Y = X^{-1}$.

(OR)

b) Show that $\sin 20° \sin 40° \sin 80° = \frac{\sqrt{3}}{8}$.

42. a) A manufacturer produces 80TV sets at a cost of ₹2,20,000 and 125 TV sets at a cost of ₹2,87,500. Assuming the cost curve to be linear, find the linear expression of the given information. Also estimate the cost of 95 TV sets.

(OR)

b) Resolve into partial fractions for the following: $\frac{x-2}{(x+2)(x-1)^2}$.

43. a) If $A + B = 45^\circ$, prove that $(1 + \tan A)(1 + \tan B) = 2$ and hence deduce the value of $\tan 22\frac{1^\circ}{2}$.

(OR)

b) $1.2 + 2.3 + 3.4 + \dots + n(n+1) = \frac{n(n+1)(n+2)}{3}$, for all $n \in N$.

44. a) If $x^m \cdot y^n = (x+y)^{m+n}$, then show that $\frac{dy}{dx} = \frac{y}{x}$.

(OR)

b) Suppose the inter-industry flow of the product of two industries are given as under.

Production Sector	Consumption sector		Domestic demand	Total output
	X	Y		
X	30	40	50	120
Y	20	10	30	60

Determine the technology matrix and test Hawkin's -Simon conditions for the viability of the system. If the domestic demand changes to 80 and 40 units respectively, what should be the gross output of each sector in order to meet the new demands.

45. a) Prove that: $\frac{\sin(180^\circ - \theta)\cos(90^\circ + \theta)\tan(270 - \theta)\cot(360 - \theta)}{\sin(360 - \theta)\cos(360 + \theta)\sin(270 - \theta)\operatorname{cosec}(-\theta)} = -1$

(OR)

b) Find the axis, vertex, focus, equation of directrix and the length of latus rectum for the parabola $x^2 + 6x - 4y + 21 = 0$

46. a) The cost of 4 kg onion, 3 kg wheat and 2 kg rice is ₹320. The cost of 2 kg onion, 4 kg wheat and 6 kg rice is ₹ 560. The cost of 6 kg onion, 2 kg wheat and 3 kg rice is ₹ 380. Find the cost of each item per kg by matrix inversion method.

(OR)

b) If $y = \sin(\log x)$, then show that $x^2 y_2 + xy_1 + y = 0$.

47. a) In how many ways can a cricket team of 11 players be chosen out of a batch of 15 players?

(i) There is no restriction on the selection.

(ii) A particular player is always chosen.

(iii) A particular player is never chosen.

(OR)

b) Find the equation of the circle passing through the points $(0,1)$, $(4,3)$ and $(1, -1)$.